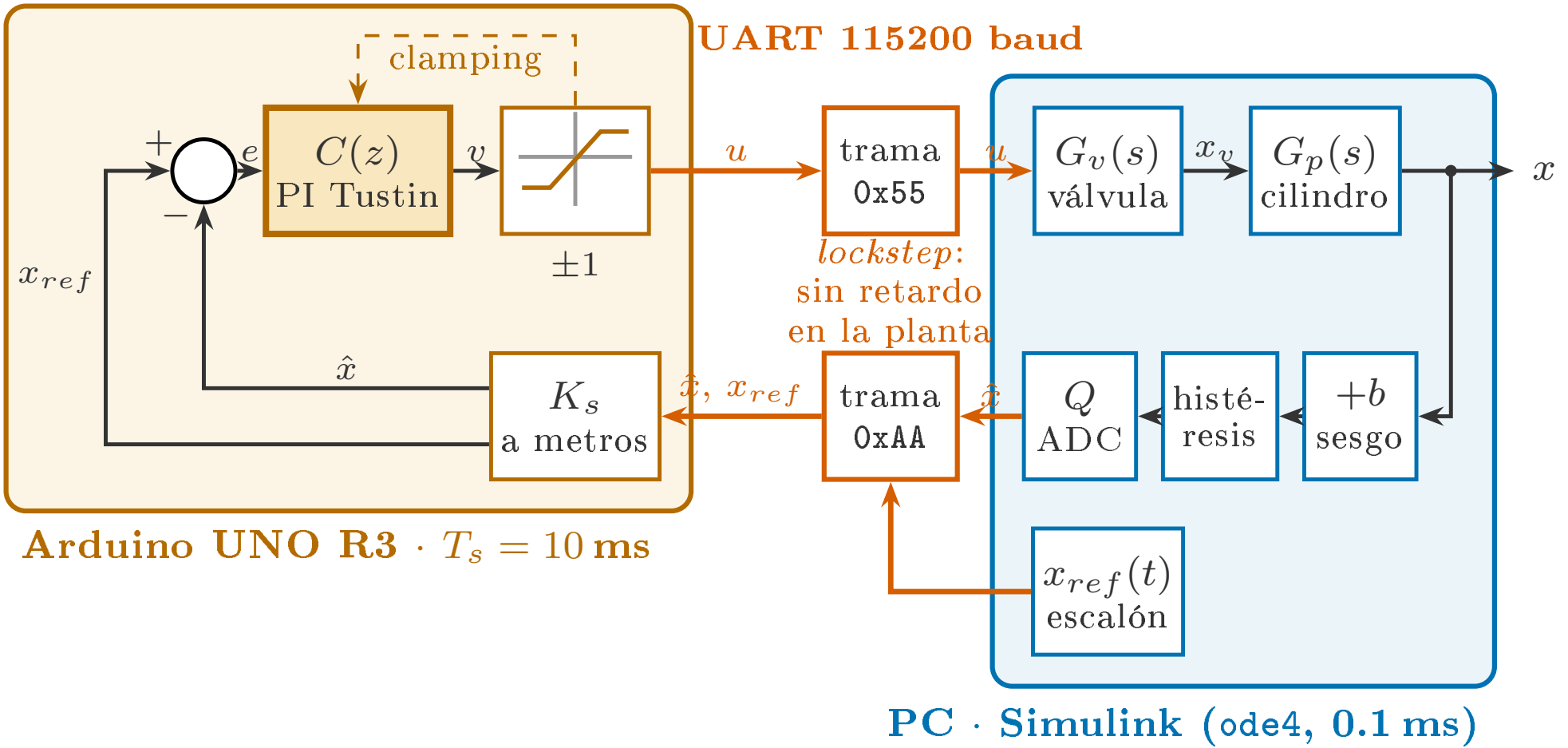

UART 115200 baud
clamping
$C(z)$
PI Tustin
v
± 1
e
$+$
$-$
x_{ref}
$\hat{x}$
K_s
a metros
u
trama
0x55
u
$G_v(s)$
válvula
x_v
$G_p(s)$
cilindro
x
lockstep:
sin retardo
en la planta
$\hat{x}$, x_{ref}
trama
0xAA
$\hat{x}$
Q
ADC
histé-
resis
$+b$
sesgo
$x_{ref}(t)$
escalón
Arduino UNO R3 · $T_s = 10$ ms
PC · Simulink (ode4, 0.1 ms)